



Analysis of optical coherence tomography angiographic findings of prematurely born children and its relationship with macular edema of prematurity

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PURPOSE	To compare foveal avascular zone (FAZ) area, foveal vascular density (VD), and foveal thickness in pre- and full-term children and to evaluate their relationship with cystoid macular edema (CME) in the prematurity period using spectral domain optical coherence tomography angiography (SD-OCTA).
METHODS	OCTA imaging was performed at 4-6 years of age in 90 eyes of 45 prematurely born children and 50 eyes of 25 term children. Subjects were divided into three groups: prematurely born with CME (group 1); prematurely born without CME (group 2); healthy, term children (group 3). Imaging results in the three groups were compared.
RESULTS	FAZ area was significantly larger in group 3 than in groups 1 and 2 ($P < 0.001$ [ANOVA]). FAZ area was found to be correlated with birth weight ($r = 0.511$; $P < 0.001$) and gestational age ($r = 0.532$; $P < 0.001$). No significant relationship was found between history of CME and FAZ area.
CONCLUSIONS	In our study cohort, FAZ area was smaller in prematurely born children and was correlated with older gestational age and higher birth weight. CME in the neonatal period did not seem to affect retinal microvascular development in premature infants. (J AAPOS 2022;26:73.e1-6)

Retinopathy of prematurity (ROP) is one of the most common causes of preventable blindness in children. Although ROP is a vascular disease, spectral domain optical coherence tomography (SD-OCT) studies have demonstrated that it causes structural changes in the fovea of children born prematurely.^{1,2} Portable SD-OCT permits visualization of macular structures in premature infants.¹⁻⁴ Macular edema is present in 30%-60% of premature babies¹⁻⁵; it is not associated with ROP stage or severity.^{1,2} On the other hand, decreased visual acuity and hypermetropia have been demonstrated in premature babies who have macular edema.³

It has been suggested that the appearance of macular edema in ROP patients is caused by vascular endothelial

growth factor (VEGF), as in retinal vein occlusion and diabetic macular edema.⁶ Mechanical traction and other neurohumoral factors are also associated with development of macular edema. Cystoid macular edema (CME) disappears spontaneously by postmenstrual age (PMA) of 52 weeks in premature babies.⁷ However, little is known about structural and microvascular changes in the maculae of prematurely born children who experienced macular edema during the neonatal period.⁸

OCT angiography (OCTA) enables a quantitative examination of the vessels of the macula and optic disk. Using fundus fluorescein angiography, Spaide and colleagues⁹ demonstrated changes in the foveal avascular zone (absence of FAZ area and smaller FAZ area) in children with a history of prematurity. However, there are few OCTA studies on ex-premature children that evaluate superficial and deep foveal vessels without fluorescein dyes.¹⁰⁻¹⁴ To our knowledge, no study examines FAZ and macular vascular density (VD) in prematurely born children with a history of CME in the neonatal period. In this study, we used OCTA to investigate FAZ area and foveal VD alterations in children 4-6 years of age who were born prematurely to assess whether CME detected during infancy was associated with subsequent abnormalities in FAZ area and foveal VD.

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Subjects and Methods

This was a single-center, observational, prospective, case-control study of prematurely born children screened for ROP in Antalya Training and Research Hospital. The medical records of all children who had undergone SD-OCT imaging during ROP screening between January 2013 and May 2015 were reviewed and these children were invited to participate in the study. OCTA imaging was performed between March 2018 and January 2019. Permission was obtained from the parents before the study to use patient information. Both eyes of participants were evaluated with OCTA. Approval for the study was obtained from the Health Sciences University Antalya Training and Research Hospital Ethics Committee and conducted in accordance with the Declaration of Helsinki.

All subjects underwent complete ocular examination including best-corrected visual acuity using Snellen chart, refraction measurement, intraocular pressure, macular SD-OCT imaging (RTVue, Optovue, Fremont, CA), axial length measurement (Lenstar, Haag-Streit, Germany) and macular OCTA imaging (RTVue, Optovue, Fremont, CA). Only one experienced examiner performed all SD-OCT and OCTA imaging on both eyes of subjects.

The main outcomes were the comparison of FAZ area, superficial and deep whole VD, superficial foveal VD (sfVD) and deep foveal VD (dfVD), parafoveal VD (pfVD), perifoveal VD (pefVD), full foveal thickness, inner foveal thickness and outer foveal thickness between prematurely born children and term children. The secondary outcome was the effect of macular edema in prematurity on retinal thickness and microvascular structure.

The inclusion criteria for prematurely born children were gestational age under 36 weeks and birth weight <2500 g and having a macular SD-OCT image taken between 36-44 weeks' PMA; for the control group, gestational age of 36 weeks or greater and

birth weight over 2500 g and normal ocular examination. Subjects with congenital eye problems or history of ocular surgery were excluded from study. A total of 49 prematurely born participants were included in the study; 4 with low-quality (SSI <60) OCTA scans were subsequently excluded. The remaining 90 eyes of 45 prematurely born subjects who were age 4-6 years were divided into two groups. The eyes diagnosed with grade 1 (20 eyes) and grade 2 (20 eyes) CME comprised group 1 (n = 40). The eyes without CME in their infancy comprised group 2 (n = 50). Eyes of 25 full term children with normal ocular examinations who came to our clinic were recruited for a control group (group 3; n = 50). The control group consisted of children of comparable age to the prematurely born children.

Macular Thickness Measurement

Macular SD-OCT imaging was previously performed in prematurely born children between postmenstrual weeks 36 and 44. Macular edema in the neonatal period was classified according to the following scale: grade 0, no edema; grade 1, macular cyst without changing macular contour; and grade 2, macular cyst with changing macular contour.⁴ Additionally highest ROP stages and ROP zone at first visit according to ICROP-revisited (International Committee of Retinopathy of Prematurity) criteria were recorded.¹⁵

The macula was divided into three concentric rings measuring 1 mm, 3 mm, and 6 mm. Full foveal thickness was defined as retinal thickness in the central 1 mm; full parafoveal thickness as retinal thickness between 1-3 mm; and full perifoveal thickness as retinal thickness between 3-6 mm. The thickness was measured by identifying the layers of the retina and measuring the distance between the internal limiting membrane and the inner boundary of the retinal pigment epithelium. Inner (from the internal limiting membrane to outer plexiform layer) and outer (from

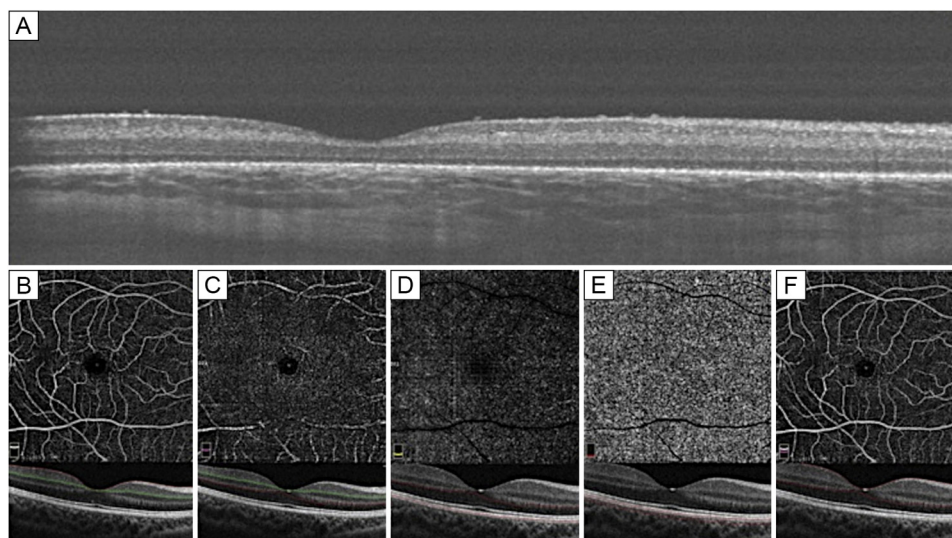


FIG 1. A premature infant without CME whose gestational age was 26 weeks and birth weight 910 g. A, Spectral domain optical coherence tomography (SD-OCT) image at postmenstrual age of 38 weeks. There are no cystoid spaces in macular SD-OCT images. The foveal contour is good. B-F, En face and structural optical coherence tomography angiography (OCTA) images. B, Superficial plexus. C, Deep plexus. D, Outer retina. E, Choriocapillaris. F, Retinal slab (internal limiting membrane to outer plexiform layer + 10 μ m).

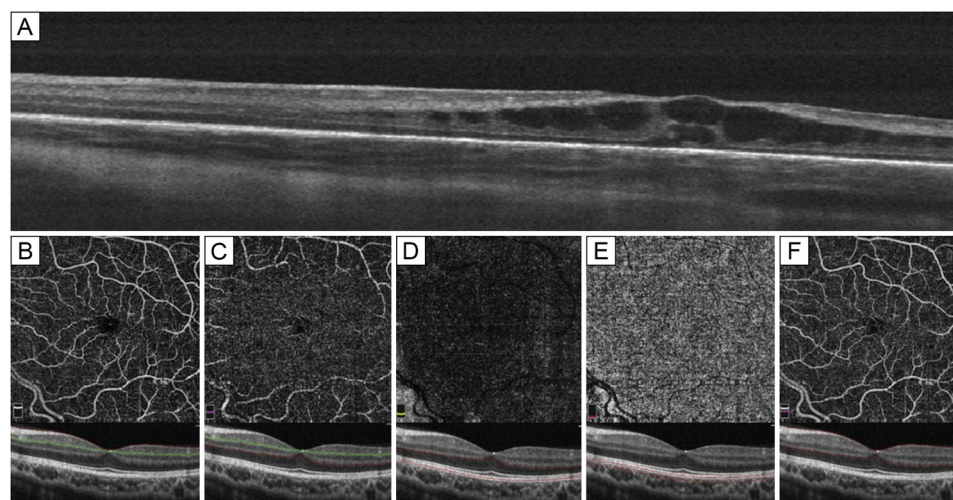


FIG 2. A premature infant with CME whose gestational age was 27 weeks and birth weight 850 g. A, SD-OCT image at postmenstrual age of 38 weeks. There are cystoid spaces in macular SD-OCT images. The foveal contour is impaired. B-F, En face and structural OCTA images. B, Superficial plexus; en face image shows the areas in which VD was measured; 1 mm foveal, 3 mm parafoveal, 6 mm perifoveal circles. C, Deep plexus. D, Outer retina. E, Choriocapillaris. F, Retinal slab (internal limiting membrane to outer plexiform layer + 10 μ m).

outer plexiform layer to retina pigment epithelium) retinal thickness were also measured separately in the same quadrants in each patient.

OCTA Measurements

The vascular structures were evaluated with 6×6 mm OCTA scans. These full-thickness retinal scans were automatically segmented into the superficial and deep inner retinal vascular plexuses, outer retina and choriocapillaris. The VD in the superficial and deep retinal vascular zone, and the FAZ area were calculated automatically by 6×6 mm OCTA software of the Optovue device. En face images of superficial VD were obtained with a slab between an inner boundary at 5.6μ m beneath the inner limiting membrane and an outer boundary at 12.6μ m beneath the inner plexiform layer (IPL) (Figure 1B, 2B). En face images of deep VD were obtained with a slab between inner and outer boundaries, respectively, at 15.6μ m and 70.2μ m beneath the IPL (Figure 1C, 2C). The FAZ area was analyzed as a single layer, and not separated into superficial and deep layers. All measurements were obtained from 9:00 and 11:00 o'clock. Only the scans which had a signal strength of a minimum of 60 were included in data analysis.

Statistical Analysis

Demographic characteristics and other data were reported in terms of the mean and standard deviation, frequency, and percentage of presence of ROP and its stage and sex. FAZ, retinal VD, grading of macular edema were analyzed using analytical software (SPSS Inc, Chicago, IL). Kruskal-Wallis and one way ANOVA (analysis of variance) Tests (Dunn Bonferoni and Tukey HSD), were used for comparisons. χ^2 test was used to compare the differences in categorical variables between groups. The Spearman correlation test was performed to assess the relationships between ordinal and numerical data. A P value of < 0.05 was considered to be statistically significant. Post hoc power analysis using G*Power was performed to estimate the power for dfVD, sVD, and FAZ. The results of post hoc power analysis showed a power of 85% for DCP-VD, 95% for SCP-VD, and 87% for FAZ.

Results

Demographic characteristics of the 45 prematurely born children (23 females [51%]) and 25 term-born children (14 females [56%]) are provided in Table 1. There was

Table 1. Patient characteristics

Demographic parameter	Group 1	Group 2	Group 3	P value ^a	P value ^{a,b}	P value ^{a,c}	P value ^{a,d}
Number of eyes	40	50	50	—	—	—	—
Female/male	10/10	13/12	14/11	0.948	—	—	—
Age at imaging, mean $\pm$ SD	5.2 ± 0.4	5.2 ± 0.6	4.8 ± 0.6	0.106	—	—	—
Birth week, mean $\pm$ SD	30.7 ± 2.3	31.1 ± 3.3	37.7 ± 1.6	<0.001	0.982	<0.001	<0.001
Birth weight, g, mean $\pm$ SD	1490 ± 409	1606 ± 450	2957 ± 243	<0.001	0.646	<0.001	<0.001

^aDunn-Bonferoni test used for comparisons; statistical significance set at $P < 0.05$.

^bDifference between group 1 (premature children with macular edema) and group 2 (premature children without macular edema).

^cDifference between groups 1 and 3 (term children).

^dDifference between groups 2 and 3.

Table 2. Comparison of FAZ area and superficial and deep VD between three groups

Variable	Group 1, mean $\pm$ SD	Group 2, mean $\pm$ SD	Group 3, mean $\pm$ SD	<i>P</i> value ^a	<i>P</i> value ^{a,b}	<i>P</i> value ^{a,c}	<i>P</i> value ^{a,d}
FAZ, mm ²	0.14 $\pm$ 0.07	0.18 $\pm$ 0.08	0.28 $\pm$ 0.1	<0.001	0.171	<0.001	0.001
Superficial parameters							
Whole VD, %	49.5 $\pm$ 2.5	50.7 $\pm$ 2.2	52.0 $\pm$ 2.0	0.004	0.065	<0.001	0.187
Foveal VD, %	27.2 $\pm$ 7.1	27.0 $\pm$ 8.3	22.8 $\pm$ 6.8	0.192	—	—	—
Parafoveal VD, %	52.8 $\pm$ 2.3	53.1 $\pm$ 2.2	55.3 $\pm$ 2.1	0.003	0.112	0.003	0.006
Perifoveal VD, %	49.6 $\pm$ 2.7	50.8 $\pm$ 2.3	52.4 $\pm$ 2.0	0.003	0.095	0.002	0.100
Deep parameters							
Whole VD, %	50.8 $\pm$ 5.3	52.5 $\pm$ 4.8	51.9 $\pm$ 4.9	0.319	—	—	—
Foveal VD, %	44.6 $\pm$ 6.0	43.8 $\pm$ 7.1	38.3 $\pm$ 9.0	0.024	0.992	0.026	0.041
Parafoveal VD, %	55.5 $\pm$ 3.5	56.9 $\pm$ 3.5	57.6 $\pm$ 2.6	0.098	—	—	—
Perifoveal VD, %	51.9 $\pm$ 5.6	54.1 $\pm$ 5.0	53.4 $\pm$ 5.5	0.184	—	—	—

FAZ, foveal avascular zone; VD, vascular density.

^aTukey HSD test used for comparisons; statistical significance set at $P < 0.05$.

^bDifference between group 1 (premature children with macular edema) and group 2 (premature children without macular edema).

^cDifference between groups 1 and 3 (term children).

^dDifference between groups 2 and 3.

no significant difference in the ages of participants at imaging time between the three subgroups ($P = 0.10$).

Twelve (30%) eyes in group 1 and 12 eyes in group 2 (24%) had spontaneous regression of ROP, while 4 eyes (13.3%) in group 1 and 6 eyes (12%) in group 2 were treated with intravitreal injection of ranibizumab. No statistical difference was found between group 1 and 2 in terms of treatment for ROP or spontaneously regressed ROP ($P = 0.235$; $P = 0.471$).

The FAZ area was significantly larger in group 3 (0.28 ± 0.12 mm²) compared to groups 1 and 2 (0.14 ± 0.07 mm², 0.18 ± 0.08 mm², resp.; $P < 0.001$). As shown in Table 2 no significant difference was found in the sfVD across the three groups ($P = 0.192$). Superficial pfVD varied by group ($P = 0.003$; higher in group 3 than groups 1 or 2). Superficial pefVD varied by group ($P = 0.003$; higher in group 3 than group 1). Deep foveal VD varied by group ($P = 0.024$; smaller in group 3 than groups 1 and 2); however, with Bonferroni correction for multiple comparisons, this finding was not statistically significant. As shown in Table 3, FAZ area was found to be correlated with gestational age ($r = 0.511$; $P < 0.001$) and birth weight, ($r = 0.532$; $P < 0.001$). sfVD was also found to be negatively correlated with birth weight ($r = -0.423$; $P < 0.001$) and gestational age ($r = -0.488$; $P < 0.001$). There was no correlation between visual acuity, axial length and FAZ area, or between visual acuity, axial length and foveal VD.

No significant difference was found between groups in full parafoveal thickness ($P = 0.057$). Full foveal thickness ($P = 0.011$) and full perifoveal thickness ($P = 0.018$) varied by group (see Table 4), although with Bonferroni correction for multiple comparisons this finding would not be considered statistically significant.

Discussion

In this study, we measured the FAZ area and foveal VD in 140 eyes of 70 children at age 4–6 years using OCTA. FAZ

area was measured using fundus fluorescein angiography by Mintz-Hittner and colleagues,¹⁰ who demonstrated the absence or small size of the FAZ area in prematurely born children and found a correlation between the FAZ area and gestational age and birth weight. Several studies have been conducted on prematurely born children using OCTA.^{11–14} To our knowledge, this is the first study to evaluate the FAZ area and foveal VD in preterm children with and without a history of CME.

Macular edema in premature infants has been reported in various ethnic groups and has been shown to resolve spontaneously.^{2–5} Vinekar and colleagues³ showed that visual acuity is worse in eyes with macular edema compared to controls through the first year of life and that these eyes had higher hyperopic refraction. However, in another study, eyes with macular edema had higher hyperopic refraction at 1 year of age, but similar refraction to those without edema at 2 years of age.¹⁶ We found no difference between eyes with macular edema and eyes without macular edema in terms of foveal VD and FAZ area at 4–6 years of age. However, this does not mean visual function of these eyes is similar to controls. Previously, Falavarjani and colleagues¹¹ demonstrated an incompleteness of the FAZ area in 42.8% of preterm children. Takagi and colleagues¹² could not measure the FAZ area in 10.4% of preterm children. In our study, by contrast, the FAZ area of all

Table 3. The correlation of variables between FAZ and superficial foveal VD

Variable	FAZ		Superficial foveal VD	
	<i>r</i>	<i>P</i> value ^a	<i>r</i>	<i>P</i> value ^a
Birth week	0.532	<0.001	−0.488	<0.001
Birth weight, g	0.511	<0.001	−0.423	<0.001
Axial length, mm	0.002	0.984	0.033	0.741
Visual acuity	−0.079	0.464	0.06	0.544

FAZ, foveal avascular zone; VD, vascular density.

^aStatistical significance set at $P < 0.05$.

Table 4. Comparison of retinal thicknesses between three groups

Parameters	Group 1, mean $\pm$ SD	Group 2, mean $\pm$ SD	Group 3, mean $\pm$ SD	P value ^a	P value ^{a,b}	P value ^{a,c}	P value ^{a,d}
FFT (μ m)	262.3 $\pm$ 15.4	257.1 $\pm$ 11.1	245.2 $\pm$ 9.95	0.011	0.505	0.008	0.065
FPFT (μ m)	313.3 $\pm$ 14.1	308.9 $\pm$ 13.7	319.0 $\pm$ 17.1	0.057	—	—	—
FPEFT (μ m)	287.1 $\pm$ 14.1	285.1 $\pm$ 14.3	297.5 $\pm$ 11.3	0.018	0.809	0.070	0.013
IFT (μ m)	78.3 $\pm$ 12.8	81.2 $\pm$ 14.5	68.2 $\pm$ 7.4	0.013	0.605	0.095	0.01
IPFT (μ m)	128.4 $\pm$ 12.8	130.7 $\pm$ 6.8	134.3 $\pm$ 6.9	0.139	—	—	—
IPEFT (μ m)	112.9 $\pm$ 12.1	115.3 $\pm$ 6.8	119.9 $\pm$ 4.6	0.948	—	—	—
OFT (μ m)	183.7 $\pm$ 8.8	176.0 $\pm$ 9.5	176.6 $\pm$ 11.2	0.002	0.002	0.083	0.979
OPFT (μ m)	184.5 $\pm$ 11.6	178.4 $\pm$ 9.6	184.6 $\pm$ 11.1	0.015	0.031	0.999	0.125
OPEFT (μ m)	173.9 $\pm$ 9.5	169.8 $\pm$ 9.9	177.7 $\pm$ 8.4	0.016	0.199	0.026	0.709

FFT, full foveal thickness; FPFT, full parafoveal thickness; FPEFT, full perifoveal thickness parafoveal; IFT, inner foveal thickness; IPFT, inner parafoveal thickness; IPEFT, inner perifoveal thickness; OFT, outer foveal thickness; OPFT, outer parafoveal thickness; OPEFT, outer perifoveal thickness.

^aTukey HSD test used for comparisons; statistical significance set at $P < 0.05$.

^bDifference between group 1 (premature children with macular edema) and group 2 (premature children without macular edema).

^cDifference between groups 1 and 3 (term children).

^dDifference between groups 2 and 3.

ex-premature children could be obtained from the OCTA scans. This might be due to a higher gestational age (30.9 weeks) and heavier birth weight (1554 g) of the participants in our study compared with these studies.

Our results indicate that prematurely born children have smaller FAZ area compared to term children, similar to Falavarjani and colleagues.¹¹ Chen and colleagues¹³ found that prematurely born patients with a history of anti-VEGF treatment for ROP had smaller FAZ area and higher foveal VD than children born at term. Nonobe and colleagues¹⁴ found smaller FAZ and abnormal pfVD in patients treated with laser photocoagulation or cryopexy compared with term children. We found the FAZ area to 0.17 mm² in our study population, slightly larger than previous studies; this may be related to differences in ROP screening criteria between countries¹⁷ or different measurement devices.

Lepore and colleagues¹⁸ assessed the structural outcome of eyes treated with intravitreally injected bevacizumab or laser photocoagulation using fluorescein angiography. They found that FAZ formation was more frequently seen in lasered eyes (64.6%) than injected eyes (25%) in 9-month-olds,¹⁸ with fewer subsequent macular abnormalities at 4 years of age.¹⁹ Young and colleagues²⁰ showed in their histopathological study that VEGF mRNA could not be detected in photocoagulated areas of the retina (although it was elevated between laser scars). On the other hand, intravitreal bevacizumab levels peak two weeks after injection and then decline.²¹ Hartnett and colleagues²² have suggested that a higher level of VEGF may cause abnormal foveal vascularization. Laser treatment may cause a more sustained decrease in VEGF and allow more normal foveal vascular development than transient suppression of VEGF by a bevacizumab injection. Lee and colleagues⁵ and Vinekar and colleagues⁷ argued that the edema at the macular region in premature infants is

due to a higher level of VEGF, as in retinal vein occlusion and diabetic macular edema. Therefore, we hypothesized that the structural and vascular development of the fovea could be predicted by macular edema in premature infants. However, we found no difference between prematurely born children with or without macular edema in infancy in terms of FAZ area and foveal VD. Takagi and colleagues¹² compared the foveal characteristics of children with a history of ROP and a control group and found higher VD at the fovea in the ROP group. On the other hand, they found lower pfVD in patients with ROP. We found that preterm children had lower superficial pfVD than children born at term and those with CME had lower superficial pefVD than children born at term. However, there were no other significant differences in VD between groups.

Our results are limited by the use of a cross-sectional study design, and by the small sample of patients from a single institution, which may limit the ability to generalize the results to a larger population. In addition, although there are different kinds of macular edema, which may differ in pathogenesis and affect foveal development in premature babies differently, we did not distinguish subtypes. Furthermore, we evaluated both eyes of each subject, which could affect results due to within-subject correlations. Another limitation is lack of SD-OCT imaging during infancy in the full-term children, although a previous study of full-term infants showed only 1 of 39 patients with a single small parafoveal cystoid structure bilaterally, unlike the multiple cystoid structures seen in premature babies.²³

In conclusion, in our study cohort, FAZ area was shown to be smaller in 4- to 6-year-olds who were born prematurely compared with term-born children. Macular edema in prematurity was not associated with FAZ area and foveal VD.

Literature Search

PubMed and Google were searched on January 18, 2021, for English-language results, using the following terms: *cystoid macular edema*, *retinopathy of prematurity*, and *OCT angiography*.

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